

Danmarks
Tekniske
Universitet



Datacenters: How Should a Future Datacenter Look Like

AUTHORS

Ignacio Díez - s253632
Anil Bhattarai - s250300
Jirui Zhu - s232173
Javier González - s253628

January 25, 2026

Contents

1	Introduction	1
2	Strategies to Improve Data Center Networks	8
2.1	Packet Switching vs. Circuit Switching	8
2.1.1	DCI network	8
2.1.2	Main issues of EPS	10
2.1.3	Main advantages of OCS	10
2.1.4	Discussion between EPS and OCS	11
2.2	Topologies	13
2.2.1	Introduction: The Spine-Leaf Bottleneck	13
2.2.2	Ring Topologies: Reducing Physical Complexity	14
2.2.3	Alternatives	16
2.3	Hybrid Edge Computing Strategies	18
2.3.1	Context	18
2.3.2	Architecture	18
2.3.3	Task offloading	19
2.3.4	Hybrid orchestration and control plane	20
2.3.5	Continuum-aware scheduling and lifecycle management	20
2.3.6	Edge-cloud interconnection (transport network)	21
3	Conclusion	22
	List of Figures	I
4	Appendix	II
	References	III

1 Introduction

Data Center (DC)

A Data Centre (DC) is a specialized facility used to store, process, and manage large volumes of digital data. It consists of critical components such as servers, storage systems, networking devices, power supply units, and cooling systems that work simultaneously to ensure continuous and reliable operation. Data centers support essential digital services including cloud computing, internet applications, online banking, communication platforms, and enterprise systems. By providing high availability, security, and scalability, data centers form the backbone of today's digital and information-driven world.

With the rapid growth of digital transformation, internet usage, and data generation, data centers have become critical infrastructure for organizations and nations. Data Redundancy, Reliability, security, scalability, and energy efficiency are the key factors that define a current data center.



Figure 1: Data Center (DC)

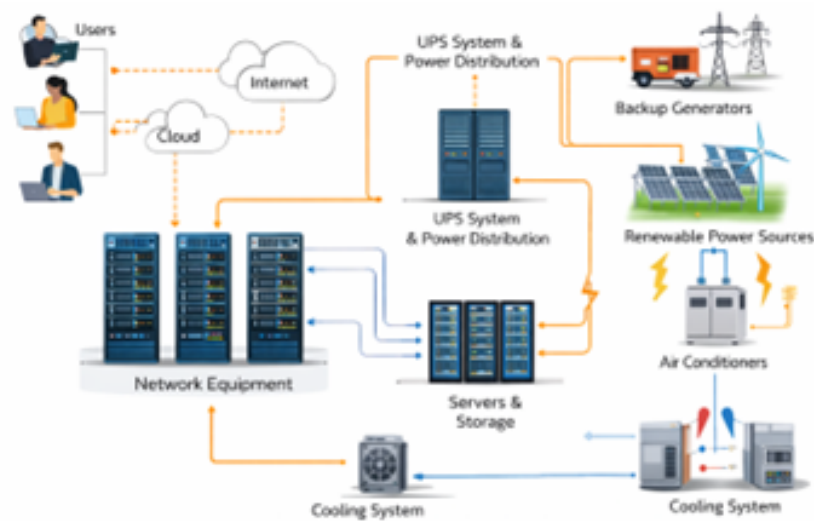


Figure 2: Infrastructure Of Data Center (DC)

Core components/infrastructure of Data Center:

Physical Infrastructure:

The physical infrastructure of a data center includes servers which can be of various types, such as rack servers, blade servers, and tower servers. Servers are used to process data across various networks. They are equipped with powerful processors, storage, memory, and can handle huge computational requirements. Server management is an essential component of a data center infrastructure to reduce cost, increase flexibility, and minimize overall downtime.

Networking Equipment's:

Networking equipment of a DC includes routers, switches, firewalls, load balancers, racks & enclosures, cabling, and fiber optics. Networking equipment of a data center enables connectivity between various components such as server storage and other external networks. Without a seamless network system, it will be impossible to deliver increased bandwidth for high-performance applications that allows organizations to manage network resources and handle traffic routing.

Storage Systems:

Storage systems can be of different types including storage area network (SAN), network attached storage (NAS), direct attached storage (DAS). Having storage system is very important to improve data management and productivity. It is also important for robot security and backup features that ensures authenticity and develops trust between operators and end-users. With the evolution of other data center components, storage systems also evolved to cater modern storage infrastructure requirements. Users also rely on cloud services for their storage needs.

Power Systems:

Power systems are an essential part of the data center infrastructure, without adequate and well-managed power supply none of the components can function. There are different types of loads in a data center including IT equipment, air conditioners, lightning, etc. The flow of energy from the generated source to the load is facilitated by various types of equipment. In this, there are two commonly used locations of equipment (i.e, the transformer upstream of the UPS) known as upstream (towards the utility unit) and downstream (towards the data center loads).

The power system of a data center infrastructure includes power distribution units (PDUs), uninterruptible power supply (UPS), backup generators, electrical panels, and power monitoring system for real-time insights on power usage and system health. The power infrastructure is designed to provide sufficient redundancy and remove all single points of failure. In instances of primary power grid faults, backup power grid can be provided to keep the system operating.

Cooling Systems:

There are various types of cooling systems such as computer air conditioning, liquid cooling, Hot and cold air flow using AC, fans and HVAC systems. Cooling and environmental control systems are important for sustainable use of power that is required to run a data center 24/7. Data centers are high density enclosed spaces that generate massive amounts of heat which cannot be contained by the traditional cooling systems, due to which there can be significant loss and increased downtime.

Security Systems:

Since many organization's trust data centers with their critical data, it is essential to have a sturdy and effective security system that covers the security aspect of a data center holistically. With the evolution of modern technology, security threats have also been evolving, resulting in a paradigm shift towards adopting a more stringent and secure data center security system.

Physical security involves surveillance cameras, biometric controlled access, and security personnel to prevent unauthorized access to the data center. On the other hand, digital security includes firewalls, intrusion detection systems, access control systems, encryption, and other software technology to ensure comprehensive security and data integrity.

Connectivity:

It can be of various types depending on the requirement and distance between the end user and the data center. For high-speed transmission, fiber optics can be used, it has high bandwidth and offers low latency. The other types of networks are peering networks and cross connects.

Its key components include switches and routers, fiber optic cables, internet service providers (ISPs), cross connects and other essential parts.

Apart from this, there are also various types of connectivity such as inter-data center connectivity (between multiple data centers), inter-data center connectivity (between servers and storage systems), and cloud connectivity (between data centers to cloud services for hybrid deployment)

Redundancy and Recovery System:

It is one of the most important systems for any data center. To avoid risks of data loss due to power outages and unanticipated natural hazards, a redundant system is required. Data centers usually offer 4 levels of redundancy, as classified by the Uptime Institute i.e, Tier 1, Tier2, Tier3, Tier4.

It includes redundant gas and power engines, waste heat recovery modules and other comprehensive cooling solutions. A data center's redundancy and disaster recovery system include various parts such as a failover system, backup solutions, and disaster recovery system for restoration of data.

Working mechanism

A data center works as a centralized facility that receives, processes, stores, and delivers digital information to users across the world. When a user sends a request through a device such as a mobile phone or computer, the request travels through the internet to the data center. Inside the data center, network devices like routers and switches direct this request to the appropriate server. The server processes the request using applications and databases, retrieves the required data from storage systems, and sends the response back to the user within seconds.

To ensure uninterrupted operation, the data center is supported by reliable power systems such as UPS and backup generators, along with efficient cooling systems that control heat generated by servers. Continuous monitoring, redundancy, and strong security measures ensure that data centers operate smoothly, securely, and without downtime.

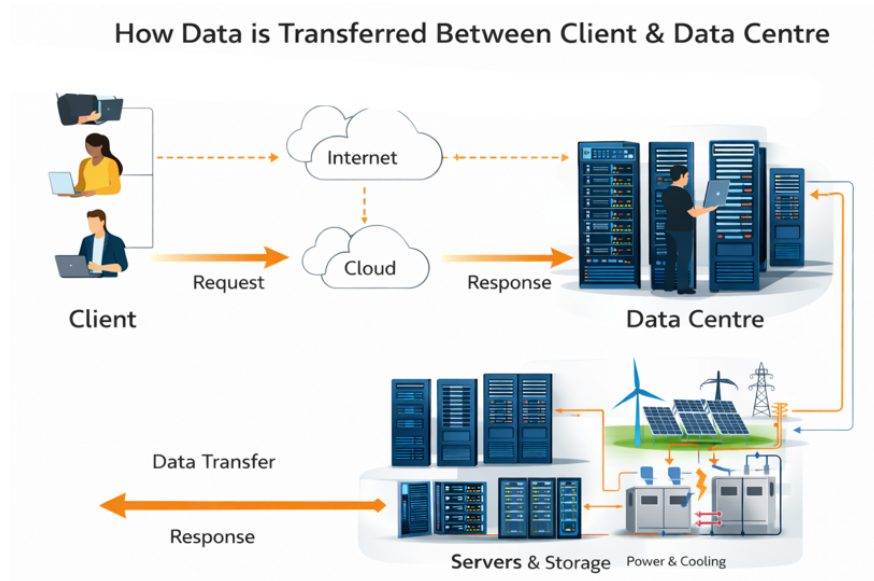


Figure 3: Working Mechanism of Data Center (DC)

Ongoing challenges on Data Center (DC):

Power and Cooling

Power and cooling continue to be primary concerns for the data center industry. According to the International Energy Agency (IEA), data centers account for about 1–1.5%

of global electricity use. To address this, data centers must focus on optimizing power usage and enhancing cooling efficiency. Implementing advanced cooling systems, such as liquid cooling solutions, can significantly reduce energy consumption in data centers compared to traditional air-based cooling methods. Additionally, the adoption of energy-efficient hardware, such as servers equipped with low-power processors, can result in significant energy savings.[1]

Scalability

Scalability remains a significant challenge for data centers as the demand for data storage and processing continues to surge. According to Grand View Research, the global data center market size was valued at \$195 billion in 2022 and is projected to grow at a compound annual growth rate (CAGR) of 10.9% from 2023 to 2030. To accommodate this growth, data centers must embrace modular and scalable designs. The use of prefabricated modules allows for rapid deployment and expansion, reducing the time and cost associated with infrastructure upgrades. Additionally, advancements in software-defined networking (SDN) and virtualization technologies enable data centers to scale their resources dynamically based on demand, ensuring efficient resource utilization. [2]

Cybersecurity

Cybersecurity has become an even more critical concern in recent years. The rise in cyber threats and the potential for data breaches have pushed data center operators to strengthen their security measures. In response, data centers must implement robust cybersecurity measures. Enhanced encryption technologies, real-time threat monitoring systems, and employee training programs can significantly reduce the risk of unauthorized access or data breaches. Furthermore, the use of artificial intelligence (AI) and machine learning algorithms can bolster data centers' ability to detect and respond to emerging threats effectively.

Data Management

Efficient data management is another challenge that data centers must tackle. IDC predicts that all the data in the world will grow to 221 zettabytes by 2026. To handle this massive volume of data, data centers must implement effective data management strategies. Advanced data classification techniques, such as machine learning-based algorithms, can automate the process of organizing and categorizing data, improving efficiency and accessibility. Additionally, the adoption of solid-state drives (SSDs) and high-density storage

solutions can optimize storage capacity and performance. Data centers must also prioritize data backup and recovery plans to ensure data integrity and quick restoration in the event of a system failure or disaster.

Sustainability

Sustainability is a pressing concern that cannot be overlooked. Recent reports indicate that data centers generate approximately 3.5% of global greenhouse gas emissions. To address this, data centers must prioritize sustainable practices. The use of server virtualization technologies has shown great potential for reducing energy consumption. Additionally, the adoption of renewable energy sources, such as solar or wind power, can further reduce the environmental impact of data centers. A report by Research and Markets says that energy-efficient data centers are expected to reach a market value of almost \$182 billion by 2026. [3]

2 Strategies to Improve Data Center Networks

2.1 Packet Switching vs. Circuit Switching

2.1.1 DCI network

In recent years, the development of cloud computing and edge services has transformed data centers from isolated computing islands into regional data center interconnection networks (DCI), consisting of multiple sites. These networks typically include multiple data centers located in the same region, which need to be interconnected at high speed to support key tasks such as cross-site data synchronization, business migration, and load balancing. As user scale and business demands increase, regional DCI needs to continuously add new data center sites and expand DCN capacity[4].

The problem is that most data centers adopt hierarchical network architectures like Clos/FatTree (Spine-Leaf), and use electrical packet switching (EPS) as the main switching/forwarding strategy. As the network scale expands, the hierarchical structure brings higher equipment and link overhead. The increasing number of network tiers will lead to a significant increase in the number of switch ports, transceivers, cabling complexity, and power consumption, thereby driving up the overall capital expenditure (CapEx) and operational expenditure (OpEx).[5]

Meanwhile, the data center traffic exhibits a strong non-uniformity and dynamics: it includes a large number of short burst flows (mice flows), as well as continuous large flows (elephant flows); and communication patterns such as incast, hotspot traffic, and data migration can cause congestion at different levels. Although EPS has mature statistical multiplexing and fine-grained scheduling capabilities, it still faces two prominent challenges:

1. **Cost and scalability** The capacity expansion of EPS-based networks typically relies on deploying additional switching nodes, ports, and high-speed transceivers (TRXs). In regional data center interconnection (Regional DCI) scenarios, studies have shown that the CAPEX of EPS scaling is largely dominated by TRX costs. As the network grows, the overall cost rises rapidly.
2. **Congestion and performance issues** Even with advanced congestion control mechanisms (including receiver-driven schemes), data centers still have problems with the coexistence of transient congestion and persistent congestion. The former often occurs at the ToR/last-hop, while the latter frequently emerges on higher-layer aggregation

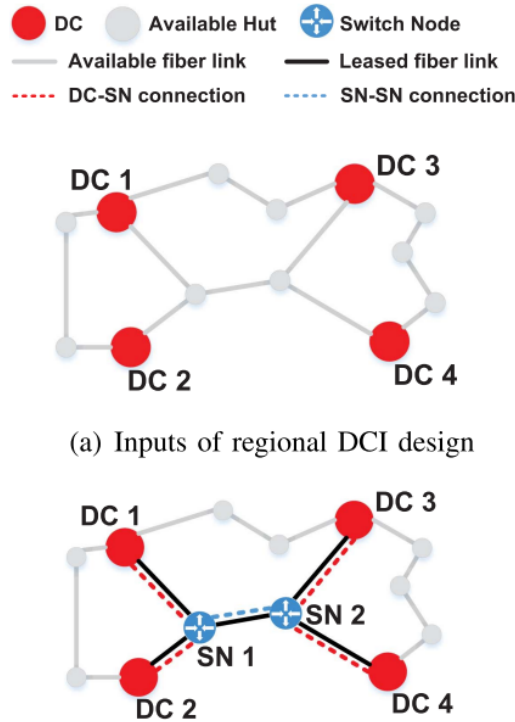


Figure 4: Example of DCI structure

links or in the core/spine. This leads to queue buildup, degraded tail latency, and reduced throughput.[6]

Under the circumstances described above, an increasing number of studies have begun to re-examine the role of optical circuit switching (OCS) in data center networks. Compared to EPS, OCS realizes data transmission by establishing end-to-end optical paths (lightpaths) to remain the switching in the link layer, which can significantly reduce hop-by-hop packet processing and queuing overheads. It can easily ensure advantages such as low latency, low power consumption, and transparent transmission. When the switching speed of circuit switching reaches magnitude of microsecond, OCS can achieve time-division multiplexing of circuit resources through rapid reconfiguration, giving it the potential to carry dynamic traffic. Moreover, the multi-level cascaded OCS network (such as COCSN) demonstrates that by utilizing WSS and multi-wavelength resources, stronger reconfigurable connectivity can be achieved, which can significantly improve network performance in hotspots or non-uniform communication scenarios.[5]

2.1.2 Main issues of EPS

In DCI network, each data center is interconnected through switching nodes (SN), and traffic is forwarded hop by hop in the form of packets. In the early stage, packet switching can fully exploit statistical multiplexing, link utilization is relatively flexible, and the deployment approach is mature. However, when DCI enters the stage of “continuous scaling” (new data centers added and bandwidth demand continuously growing), EPS expansion will reveal significant cost and structural issues.

1. **CAPEX Dominated by TRX and Ports:** In the EPS-based expansion CAPEX, the cost of high-speed transceivers (TRX) is the most significant contributor. EPS expansion presents an "inevitable linear stacking" pattern depending on the rate of the transceiver and the number of ports, which makes a burst increase in CAPEX.
2. **New SN added subject to the constraints of distance and latency:** The data forwarding between DCs requires strict control over latency. In DCI network under EPS strategy, the new data center added may not be directly connected to the existing switching nodes due to the length constraint. Therefore, additional SN devices are often required to ensure the feasibility of the connection. This leads to more nodes introduced to the network and make the DCN topology even more complicated. The new SN will also require additional investment in ports and equipment thus increasing the CAPEX. [7]
3. **The congestion and queuing in packet switching challenges the stability of performance:** In the case of high load, high traffic density or sudden business operations, the EPS network is prone to queue congestion and overloading, which in turn leads to deterioration of tail latency and other performance issues.

2.1.3 Main advantages of OCS

The basic idea of OCS is to establish end-to-end optical paths between data centers, keeping the traffic in link layer, thereby supporting high-bandwidth low-latency interconnection with lower equipment costs. Unlike EPS's "fine-grained packet statistical multiplexing", the OCS has more advantage in the hardware cost and the switching method.

Unlike EPS, the cost of OCS expansion merely involves an increase in the demand for additional fibers. Thus, OCS can significantly reduce the reliance on TRX and EPS ports, lowering the cost of capacity expansion.[4].

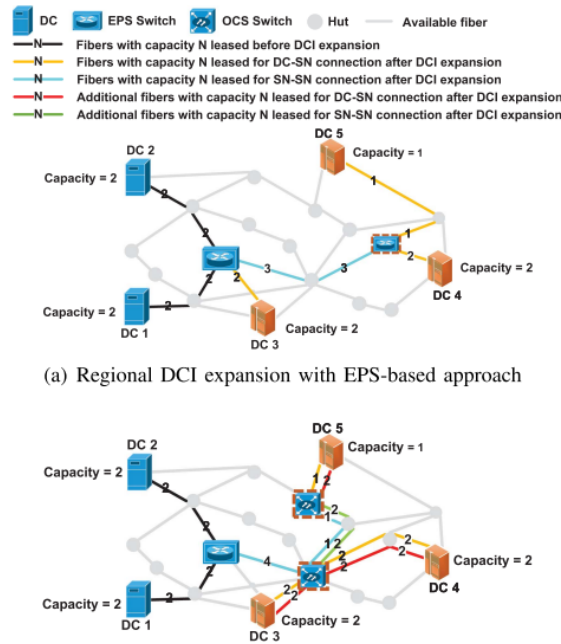


Figure 5: Example of OCS based DCI network

Moreover, in DCI network, cross-site traffic usually includes tasks such as data synchronization, backup, migration, and scheduling. These types of services often have more obvious "continuity and large bandwidth" characteristics. For such communications, hop-by-hop queuing and packet processing in packet switching are not always necessary; instead, they bring additional overhead. OCS can reduce processing at intermediate nodes by establishing circuit direct connections, and logically is more suitable for carrying such traffic profiles.[5]

In the future, if circuit switching can reach microsecond level (11.5 s), circuits can be rapidly scheduled according to the traffic matrix. This capability enables OCS to evolve from static circuits to time-division-multiplexed, programmable circuit resources. Once such fast switching is achieved, OCS not only reduces deployment and operational costs, but also approaches packet-switched networks in terms of performance and flexibility. As a result, the attractiveness of OCS for future DCI expansion is further enhanced.

2.1.4 Discussion between EPS and OCS

In summary, the main challenge of EPS in the expansion of Regional DCI lies in the cost structure is inevitably driven by TRX and ports. Unavoidable additional switching nodes introduced also results in a rapid increase in CAPEX as the scale grows. While OCS

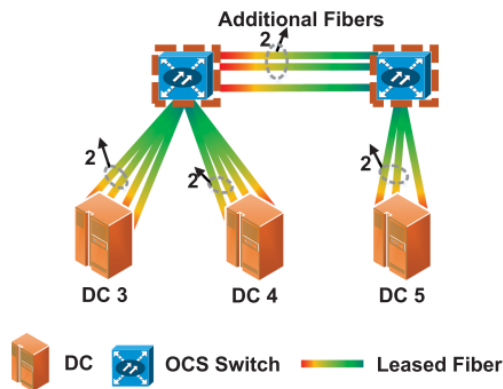


Figure 6: OCS DCI network, only fiber added when expanding the capacity

can significantly reduce TRX and SN cost and avoid topology complexity, its scheduling complexity, reconfiguration overhead, and the adaptability to short burst traffic determine that the OCS is more suitable to serve as a supplement to EPS or to carry specific services, rather than simply replacing EPS in DCN.

2.2 Topologies

2.2.1 Introduction: The Spine-Leaf Bottleneck

The architecture of data center networks is currently undergoing a fundamental paradigm shift. For the past decade, the *Spine-Leaf* topology, based on the non-blocking folded-Clos network design, has served as the backbone of the cloud computing era. By flattening the traditional three-tier hierarchy (Core, Aggregation, Access) into two functionally distinct layers, Spine-Leaf offered a standardized method to scale capacity while maintaining predictable latency. However, as the industry transitions towards exascale computing driven by Artificial Intelligence (AI) and Large Language Model (LLM) training, the limitations of this hierarchy have become a critical bottleneck.

The primary design philosophy of Spine-Leaf is connectivity: every Leaf switch (Top-of-Rack) connects to every Spine switch. While this design is great at handling the client-server traffic ("North-South") typical of the web 2.0 era, it struggles with the massive "East-West" traffic patterns generated by distributed machine learning. In these modern workloads, servers communicate constantly with one another to synchronize parameters (all-reduce operations), requiring massive, sustained bisection bandwidth. To support this without blocking, a Spine-Leaf network requires a full-mesh configuration that becomes really expensive.

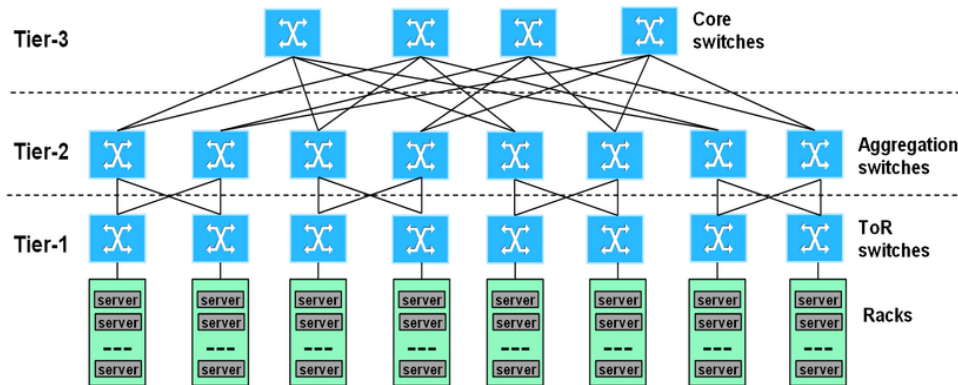


Figure 7: Visual representation of the Spine-Leaf hierarchy.

A critical inefficiency of this model lies in its lack of scalability. As demonstrated by Singla et al. [8], the structural rigidity of Clos networks imposes a “step-function” cost model. To expand the network capacity slightly—for example, to add just a few racks—an operator often cannot simply add a few cables. Instead, they must deploy entire additional Spine switches to maintain non-blocking performance across the fabric. Singla’s research

highlights that this lack of “incremental expansion” forces organizations to over-provision hardware significantly, resulting in massive capital expenditure (CAPEX) inefficiencies.

Beyond economics, the physical constraints of Spine-Leaf are reaching their breaking point. The requirement to maintain a Full Mesh results in a cabling complexity of approximately $O(N^2)$. In a large data center with thousands of servers, this manifests as unmanageable cable bundles that obstruct cooling airflow and complicate hardware maintenance. Also, the reliance on electronic packet switching at the Core (Spine) layer introduces a severe power penalty. Every packet traversing the data center must be converted from optical to electrical signals, processed, and reconverted at the Spine. This Optical-Electrical-Optical (OEO) conversion consumes a disproportionate amount of energy for traffic that is simply passing through, necessitating the exploration of flatter, optically-enabled topologies.

2.2.2 Ring Topologies: Reducing Physical Complexity

While the Spine-Leaf architecture provides excellent bisection bandwidth, it imposes a severe physical penalty: the explosion of cabling and transceiver costs. As noted in the previous section, maintaining a non-blocking full-mesh requires the number of cables to grow quadratically relative to the number of switches. To address this problem Ring topologies offer a radical geometric alternative by minimizing the physical degree of the network.

The High-Radix Problem In a traditional Clos network (Spine-Leaf), switches are "High-Radix," meaning they require a large number of ports (e.g., 64, 128 ports) to connect to the layer above. A significant portion of the data center's power and budget is consumed not by the computation, but by the optical transceivers required to light up these thousands of ports. Ring architectures challenge this premise by utilizing "Low-Radix" connectivity. In a pure ring topology, a rack theoretically needs connection only to its immediate left and right neighbors. This drastic reduction in port count shifts the complexity from the physical layer (cables) to the time domain (scheduling).

RotorNet: The Dynamic Ring A static ring is unsuitable for modern data centers because the "hop count" (diameter) is too high; sending data from Node 1 to Node 50 would require passing through 49 intermediate switches, causing unacceptable latency. To solve this, Mellette et al. proposed **RotorNet** [9], which implements a *dynamic* ring topology. Instead of fixed static cables, the connections between racks rotate periodically.

Conceptually, RotorNet decouples the physical cabling from the logical topology.

1. **Physical Layer:** Racks are connected to a central passive cyclic coupler. This creates a physical loop where light can be directed.
2. **Logical Rotation:** The topology cycles through a series of permutations. In timeslot t_1 , Rack A connects to Rack B. In timeslot t_2 , Rack A connects to Rack C.

Over a complete cycle (typically hundreds of microseconds), every rack gets a direct opportunity to transmit to every other rack. This effectively emulates a Full Mesh topology over time, but uses only a fraction of the transceivers required by a Spine-Leaf architecture.

Cost and Scalability Analysis The primary advantage of the Ring approach is the linearization of infrastructure costs.

- **Spine-Leaf Cost:** $O(N \times R)$, where N is the number of racks and R is the radix of the spine switches. Expanding the network often requires purchasing entire new spine chassis ("step function" cost).
- **Ring Cost:** $O(N)$. Adding a new rack to a ring topology simply involves inserting it into the loop (connecting it to its neighbors).

Research indicates that for exascale data centers, ring-based dynamic topologies can reduce the total number of required optical transceivers by nearly 60% compared to a Fat-Tree network of equivalent throughput. Since transceivers account for a major portion of the network CAPEX, this represents a massive economic advantage for hyperscalers.

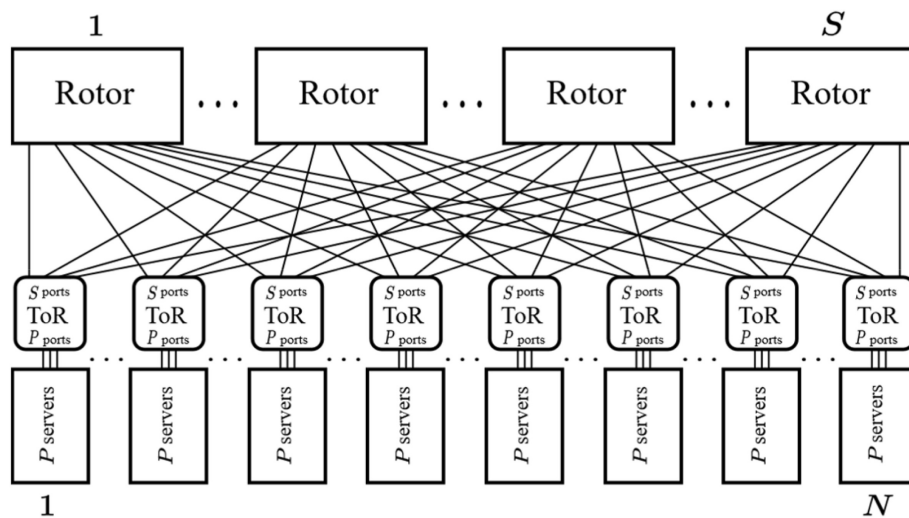


Figure 8: Structure of the RotorNet topology.

Latency and Buffering Challenges The trade-off for this physical simplicity is latency management. In a Spine-Leaf, a path always exists. In a dynamic Ring like RotorNet, a packet arriving at a switch might have to "wait" until the ring rotates to the correct position for the destination rack. This introduces a "Store-and-Forward" behavior that differs from the "Cut-Through" switching of traditional Ethernet fabrics. To mitigate this, modern ring architectures employ sophisticated buffering strategies and cycle synchronization to ensure that the average wait time remains within the microsecond range, acceptable for most batch-processing and AI-training workloads, though potentially challenging for ultra-low-latency financial trading applications.

In conclusion, Ring topologies represent a shift from spatial complexity (more cables) to temporal complexity (smart scheduling), offering a sustainable path for scaling data centers beyond the physical limits of the Spine-Leaf model.

no

2.2.3 Alternatives

While the previous section highlighted the physical limitations of hierarchical trees, a parallel stream of research focuses on optimizing the logical graph structure. Traditional architectures like Fat-Tree or even early low-diameter attempts like Dragonfly (2008) suffer from rigid scalability rules—adding capacity often requires extensive rewiring or massive upfront capital investment. To address the dynamic nature of Exascale and AI workloads, recent literature proposes graph-based topologies that prioritize distinct flexibility and specific traffic patterns.

PolarFly: Flexible Low-Diameter Networks

A major breakthrough in topology design is PolarFly, introduced by Lakhotia et al. [10]. It represents the first diameter-2 topology that approaches the theoretical Moore bound while allowing for *incremental scalability*.

Previous low-diameter topologies (like Slim Fly) were statically defined; growing the cluster meant replacing the entire graph structure. PolarFly leverages the mathematical properties of Erdős-Rényi polarity graphs to create a modular design. Its key innovation lies in its ability to support the addition of single routers without global rewiring. This "incremental growth" capability solves the main economic drawback of Spine-Leaf (the step-function cost), allowing data centers to scale organically node-by-node rather than block-by-block. Furthermore, by ensuring a maximum of two hops between any endpoints, PolarFly minimizes tail latency for synchronization-heavy workloads.

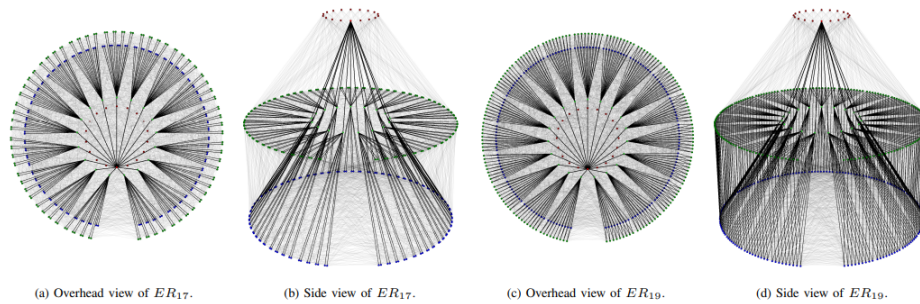


Figure 9: Visual representation of a low-diameter graph topology.

HammingMesh: Optimized for Deep Learning While PolarFly optimizes for general-purpose low latency, HammingMesh (Hoeffler et al., 2023) [11] is designed specifically for the traffic patterns of Large Language Model (LLM) training and Deep Learning.

Modern AI workloads exhibit a distinct dichotomy in bandwidth demand: they require extreme bandwidth between local GPUs for model parallelism and moderate bandwidth for global gradients (data parallelism). HammingMesh exploits this by distinguishing between local and global dimensions. It utilizes high-density 2D grids for neighbor-to-neighbor communication (using cost-effective cabling) and a sparse flexible global layer for long-distance synchronization.

This hybrid approach decouples local geometry from global topology, allowing data centers to scale "out" (adding more racks) and "up" (adding more GPUs per rack) independently. Hoeffler's research demonstrates that this topology achieves near-optimal performance for All-Reduce operations—the most common communication pattern in AI training—at a fraction of the cost and cabling complexity of a full Fat-Tree.

2.3 Hybrid Edge Computing Strategies

2.3.1 Context

Edge computing is a distributed computing paradigm that places computation and storage closer to where data is generated and consumed, enabling faster response times for delay-sensitive services. [12, 13] Rather than replacing cloud computing, the edge typically complements it: latency-critical processing can be performed near users, while the cloud remains essential for large-scale aggregation, centralized management, and deeper analytics. [12]

From the perspective of data center evolution, this implies that compute and storage are no longer confined to centralized facilities but are increasingly distributed across a device-to-cloud continuum, where different tiers collectively form a logically unified data center.

This shift is driven by the growth of data-intensive and real-time applications (e.g., IoT monitoring, connected and autonomous systems, and healthcare), where sending all raw data to remote data centers can create network congestion and unacceptable end-to-end delays. [12, 13] In practice, the term *edge* covers several closely related deployment models, including fog computing, cloudlets, and multi-access edge computing (MEC), which differ mainly in where resources are placed across the device-network-cloud continuum. [13]

Despite its benefits, edge computing introduces new constraints compared to hyperscale clouds. Edge data centers are geographically distributed and often deployed in non-ideal environments (e.g., urban locations), which can reduce energy efficiency and make cooling and automation more challenging. [12] In addition, local demand can be highly variable due to user mobility and handovers, making it inefficient to provision every site for peak load and motivating collaboration among edge sites and with the cloud. [12]

2.3.2 Architecture

While edge computing mitigates the latency limitations of centralized clouds by bringing cloud-like capabilities closer to end devices, edge resources are typically more constrained and heterogeneous than the practically elastic capacity of large cloud data centers. [14] As a result, many emerging services are best supported by *hybrid* edge-cloud computing, where multiple edge sites collaborate with a central cloud to form a device-to-cloud continuum and dynamically place computation across tiers. [14]

From an architectural viewpoint, the hybrid model introduces an additional resource layer between end devices and the cloud, enabling the execution of latency-sensitive components near users while still leveraging the cloud for overflow capacity and centralized

processing. [14] Concretely, end devices connect through the access network to nearby edge servers (e.g., at gateways, base stations, or micro data centers), and edge sites may further rely on the transport network to reach centralized cloud resources when local execution is insufficient. [14]

This layered organization (device–access–edge–transport–cloud) can be seen as a logical decomposition of a future data center into geographically distributed tiers. It provides the substrate for mechanisms such as computation *task offloading*, where tasks can be executed locally, at the edge, or at the cloud depending on application requirements and resource availability. [14] In hybrid settings, offloading decisions are inherently hierarchical: devices decide what to offload, while the edge further decides what remains local and what must be forwarded to the cloud, accounting for the additional delay over the edge–cloud interconnection. [14]

2.3.3 Task offloading

Task offloading refers to the transfer of resource-intensive computational tasks from end devices to an external, resource-rich platform, typically located at the edge and/or the cloud. [14] Offloading can accelerate latency-sensitive applications and improve energy efficiency for battery-powered devices, provided that the communication overhead introduced by the network is properly accounted for. [14]

From a workflow perspective, offloading is a multi-stage process that includes (i) application partitioning, (ii) offloading decision making, and (iii) distributed task execution. [14] The end device selects which tasks are executed locally and which are transmitted through the wireless access network towards an edge or cloud server for remote execution; the results of local and remote execution are then combined to produce the application output. [14]

A key classification criterion is the *granularity* of offloading. [14] In *partial offloading at the edge*, only a subset of tasks is executed remotely while the rest remains local, improving performance at the cost of more complex scheduling. [14] In *full offloading at the edge*, the entire workload is executed remotely, maximizing device-side energy savings but still requiring careful management of transmission power and QoS/QoE constraints. [14]

Hybrid deployments further enable *edge–cloud collaboration*, where offloaded tasks may be executed partly at the edge and partly at the cloud when edge resources are insufficient. [14] This hierarchical offloading process requires a control plane that can deploy, observe, and reconfigure services across heterogeneous edge and cloud resources. [14, 15] Because load and connectivity vary over time, hybrid execution depends on continuous visibility of node and network conditions, making offloading part of a broader orchestration loop that

manages placement, scaling, and re-placement across the continuum. [15]

2.3.4 Hybrid orchestration and control plane

A practical hybrid edge–cloud platform typically relies on a cloud-like orchestrator (e.g., Kubernetes) as a baseline control plane, extended with continuum-aware mechanisms, to deploy services, enforce policies, and track runtime state across distributed resources. [15] Orchestration mechanisms designed for centralized clouds therefore require extensions to handle edge constraints such as resource scarcity, strong timing requirements, and the importance of location and network proximity. [15]

The control plane closes the loop between decisions and execution by receiving deployment intents, collecting telemetry about compute and network conditions, and actuating placement, scaling, and re-placement actions. [15] This feedback loop is essential in hybrid environments, where edge nodes may become saturated or connectivity may fluctuate, making static placement decisions sub-optimal. [15]

Recent work extends Kubernetes with richer service descriptors, scheduling plugins that account for dependency latency and real-time feasibility, continuous monitoring, and latency-oriented load balancing. [15]

KubeEdge exemplifies a Kubernetes-native approach, separating cloud-side control (Cloud-Core) from edge-side execution (EdgeCore) to support geographically distributed nodes under intermittent connectivity. [16] Such platforms provide the foundation of hybrid control planes, but additional continuum-aware mechanisms are required to meet stringent edge requirements. [15]

2.3.5 Continuum-aware scheduling and lifecycle management

In the edge–cloud continuum, orchestration must be treated as a continuous control loop rather than a one-shot decision, since both resource availability and network conditions evolve over time. [15] This is particularly important for critical services with strict response-time requirements co-located with best-effort workloads on constrained edge nodes. [15]

Service descriptors must therefore capture edge-specific constraints such as location and inter-component dependencies, enabling latency-aware placement across the service chain. [15] Continuum-aware scheduling extends Kubernetes with dependency- and latency-aware placement and real-time prioritization, complemented by runtime enforcement of execution priorities on worker nodes. [15]

Lifecycle management closes the loop through monitoring and reconfiguration: when deployments no longer meet objectives, the system can trigger eviction and re-scheduling.

[15] Request routing must align with placement by steering traffic to replicas that minimize response time using network-latency and application signals. [15]

2.3.6 Edge–cloud interconnection (transport network)

Hybrid execution requires predictable connectivity between distributed continuum nodes, since microservices are often split across edge and cloud sites. [17] Control-plane connectivity can rely on best-effort Internet or VPNs, whereas data-plane interconnection must satisfy application-driven requirements for latency, bandwidth, loss, and resiliency. [17]

Because edge and cloud sites may belong to different providers, direct control of the underlay is limited, motivating overlay interconnections across heterogeneous WAN links. [17] SD-WAN provides such overlays with policy-driven, application-aware routing. [17]

Through northbound APIs, SD-WAN controllers can expose connectivity abstractions to orchestrators, enabling SLA-aware path selection and dynamic rerouting under QoS degradation. [17] This network programmability complements continuum-aware scheduling by jointly adapting compute placement and inter-site paths to preserve end-to-end performance. [17]

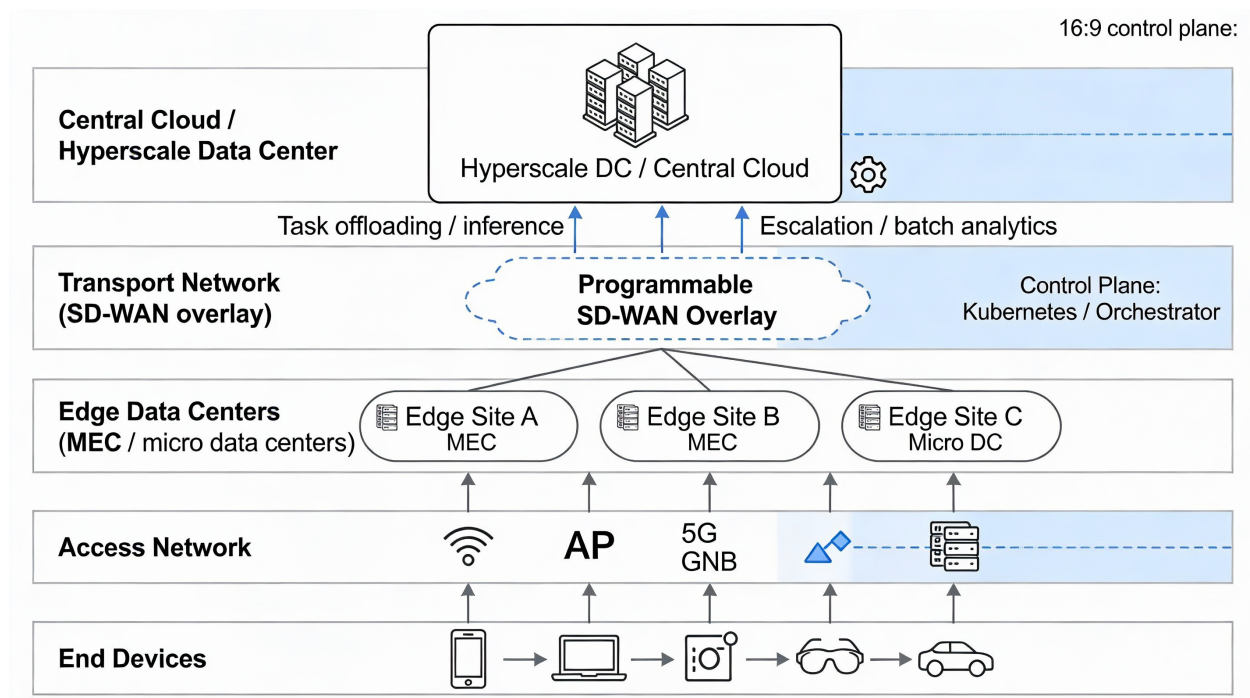


Figure 10: Edge–cloud continuum architecture for future data centers

3 Conclusion

The research and analysis presented in this document highlight that the data center industry is currently going through an inflection point. The traditional architectural models that powered the cloud computing era of the last decade—characterized by hierarchical Spine-Leaf topologies, electrical packet switching, and centralized processing—are approaching their physical and economic limits. As the demand for Exascale computing, driven largely by Artificial Intelligence (AI) training and distributed Machine Learning workloads, continues to grow exponentially, the "future data center" must evolve from a static storage facility into a dynamic, optically-enabled, and geographically distributed continuum.

The Shift from Electrical to Optical Switching

A fundamental finding of this review is the diminishing return of Electrical Packet Switching (EPS) in the context of Data Center Interconnection (DCI). While EPS has served as the standard for fine-grained statistical multiplexing, its reliance on Optical-Electrical-Optical (OEO) conversion at every hop has created an unsustainable trajectory for power consumption and capital expenditure (CAPEX). As noted in the analysis of regional DCI expansion, the cost of scaling EPS networks is dominated by the linear accumulation of transceivers and ports.

Consequently, the adoption of Optical Circuit Switching (OCS) emerges not merely as an alternative, but as a necessity for handling "elephant flows" and massive data synchronization tasks. By maintaining data in the optical domain and utilizing technologies such as MEMS-based mirrors or tunable lasers, operators can decouple bandwidth growth from power consumption. However, the future architecture will likely remain hybrid in the near term; while OCS offers superior efficiency for bulk data transport, the reconfiguration latency challenges identified in pure optical networks suggest that EPS will continue to play a role for latency-sensitive, short-lived "mice flows" until microsecond-scale optical switching becomes commercially ubiquitous.

Breaking the Topological Rigidity

Perhaps the most significant architectural shift identified is the transition from rigid hierarchical trees to flexible graph-based topologies. The Spine-Leaf architecture, despite its predictability, suffers from a lack of incremental scalability ("step-function" costs) and creates cabling complexities of $O(N^2)$ that obstruct physical airflow and maintenance.

The exploration of alternative topologies reveals two distinct paths forward. On the physical layer, Ring Topologies like RotorNet demonstrate that shifting complexity from the spatial domain (cabling) to the temporal domain (time-division multiplexing) can reduce transceiver requirements by nearly 60%. This represents a massive potential for cost reduction in hyperscale facilities. On the logical layer, the emergence of low-diameter graphs such as PolarFly and HammingMesh validates the need for topologies designed specifically for modern traffic patterns. Unlike general-purpose trees, these architectures leverage mathematical properties (such as Erdős-Rényi graphs) to ensure low hop counts and support the distinct "East-West" bandwidth demands of parallel AI training. Crucially, they solve the economic bottleneck of Spine-Leaf by enabling granular, node-by-node expansion.

The Device-to-Cloud Continuum

Finally, the concept of the data center has expanded beyond the physical walls of a centralized facility. The integration of Hybrid Edge Computing strategies redefines the data center as a logical continuum stretching from end-user devices to the hyperscale cloud. The analysis of task offloading and orchestration platforms like KubeEdge illustrates that future performance will not solely depend on raw core capacity, but on the intelligent placement of workloads.

By decoupling the local geometry of interconnections from the global topology, and by implementing continuum-aware scheduling, future infrastructures can mitigate the latency limitations of the speed of light. The challenge moving forward lies in the control plane: orchestrating this heterogeneous environment requires sophisticated algorithms capable of balancing energy efficiency, latency constraints, and dynamic network conditions in real-time.

In summary, the future data center will be defined by its plasticity. It will rely on optical backbones to minimize energy waste, utilize graph-based topologies to maximize incremental scalability, and leverage edge-cloud collaboration to ensure proximity to the user. Achieving this vision requires a holistic abandonment of the rigid legacy structures of the past in favor of flexible, software-defined, and physically optimized designs.

List of Figures

1	Data Center (DC)	1
2	Infrastructure Of Data Center (DC)	2
3	Working Mechanism of Data Center (DC)	5
4	Example of DCI structure	9
5	Example of OCS based DCI network	11
6	OCS DCI network, only fiber added when expanding the capacity	12
7	Visual representation of the Spine-Leaf hierarchy.	13
8	Structure of the RotorNet topology.	15
9	Visual representation of a low-diameter graph topology.	17
10	Edge-cloud continuum architecture for future data centers	21

4 Appendix

Anil is Responsible for **Introduction** part.

Jirui is responsible for **Switching Technology** part.

Ignacio is responsible for **Topology** and **Conclusion** part

Javior is responsible for **Edge Computing** part.

References

- [1] P. James Parker, C. CEM, P. Bill Brown, and H. Lindsay, “Tackling today’s data center energy efficiency challenges,” 2012.
- [2] T. Murino, R. Monaco, P. S. Nielsen, X. Liu, G. Esposito, and C. Scognamiglio, “Sustainable energy data centres: A holistic conceptual framework for design and operations,” *Energies*, vol. 16, no. 15, 2023.
- [3] A. Radovanović, R. Koningstein, I. Schneider, B. Chen, A. Duarte, B. Roy, D. Xiao, M. Haridasan, P. Hung, N. Care, S. Talukdar, E. Mullen, K. Smith, M. Cottman, and W. Cirne, “Carbon-aware computing for datacenters,” *IEEE Transactions on Power Systems*, vol. 38, no. 2, pp. 1270–1280, 2023.
- [4] X. Yu *et al.*, “Network expansion of regional dcis: Optical circuit switching versus electrical packet switching,” *Journal of Lightwave Technology*, vol. 42, no. 20, pp. 7044–7055, 2024.
- [5] S. Li *et al.*, “Cocsn: A multi-tiered cascaded optical circuit switching network for data center,” *IEEE Transactions on Cloud Computing*, vol. 12, no. 4, pp. 1463–1475, 2024.
- [6] Q. Huang *et al.*, “Combining circuit and packet switching using a large port-count optical cross-connect for data center networks,” *Optics Communications*, vol. 285, no. 21–22, pp. 4268–4274, 2012.
- [7] X. Zhang *et al.*, “A receiver-driven transport protocol using differentiated algorithms for differential congestion in datacenters,” *Computer Networks*, vol. 245, p. 110357, 2024.
- [8] A. Singla, C.-Y. Hong, L. Popa, and P. B. Godfrey, “Jellyfish: Networking data centers randomly,” *USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, pp. 225–238, 2012.
- [9] W. M. Mellette, R. McGuinness, A. Roy, G. Porter, and A. C. Snoeren, “Rotornet: A scalable, low-complexity, optical datacenter network,” in *Proceedings of the Conference of the ACM Special Interest Group on Data Communication (SIGCOMM)*, pp. 225–238, 2017.
- [10] K. Lakhotia, M. Besta, L. Monroe, K. Isham, P. Iff, T. Hoefer, and F. Petrini, “Polarfly: A cost-effective and scalable low-diameter topology,” in *ISCA*, 2022.

-
- [11] T. Hoeffler, T. Bonato, D. De Sensi, S. Di Girolamo, S. Li, M. Heddes, J. Belk, and D. Goel, “Hammingmesh: A network topology for large-scale deep learning,” *ACM Transactions on Architecture and Code Optimization (TACO)*, 2023.
- [12] P. Arroba, R. Buyya, R. Cárdenas, J. L. Risco-Martín, and J. M. Moya, “Sustainable edge computing: Challenges and future directions,” *Software: Practice and Experience*, 2024.
- [13] W. Z. Khan, E. Ahmed, S. Hakak, I. Yaqoob, and A. Ahmed, “Edge computing: A survey,” *Future Generation Computer Systems*, vol. 97, pp. 219–235, 2019.
- [14] F. Saeik, M. Avgeris, D. Spatharakis, N. Santi, D. Dechouniotis, J. Violos, A. Leivadeas, N. Athanasopoulos, N. Mitton, and S. Papavassiliou, “Task offloading in edge and cloud computing: A survey on mathematical, artificial intelligence and control theory solutions,” *Computer Networks*, vol. 195, p. 108177, 2021.
- [15] R. Rosmaninho, D. Raposo, P. Rito, and S. Sargento, “Edge-cloud continuum orchestration of critical services: A smart-city approach,” *IEEE Transactions on Services Computing*, 2025. Also available as arXiv:2407.17314.
- [16] S.-H. Kim and T. Kim, “Local scheduling in kubeedge-based edge computing environment,” *Sensors*, vol. 23, no. 3, p. 1522, 2023.
- [17] A. Mokhtari and A. Ksentini, “Sd-wan for cloud edge computing continuum interconnection,” in *2024 IEEE Global Communications Conference (GLOBECOM)*, (Cape Town, South Africa), Dec. 2024. Available: <https://www.eurecom.edu/publication/7823/download/comsys-publi-7823.pdf>.